

Experiment 7

Implementation of Elliptic Curve Cryptography (ECC)

Aim

To implement a simple Elliptic Curve Cryptography (ECC) algorithm for generating a shared secret key between two users.

Description

Elliptic Curve Cryptography (ECC) is a public-key cryptographic technique based on the mathematics of elliptic curves. It provides the same level of security as RSA while using much smaller key sizes, making it faster and more efficient. ECC is widely used in secure communication, digital signatures, and key exchange.

Algorithm

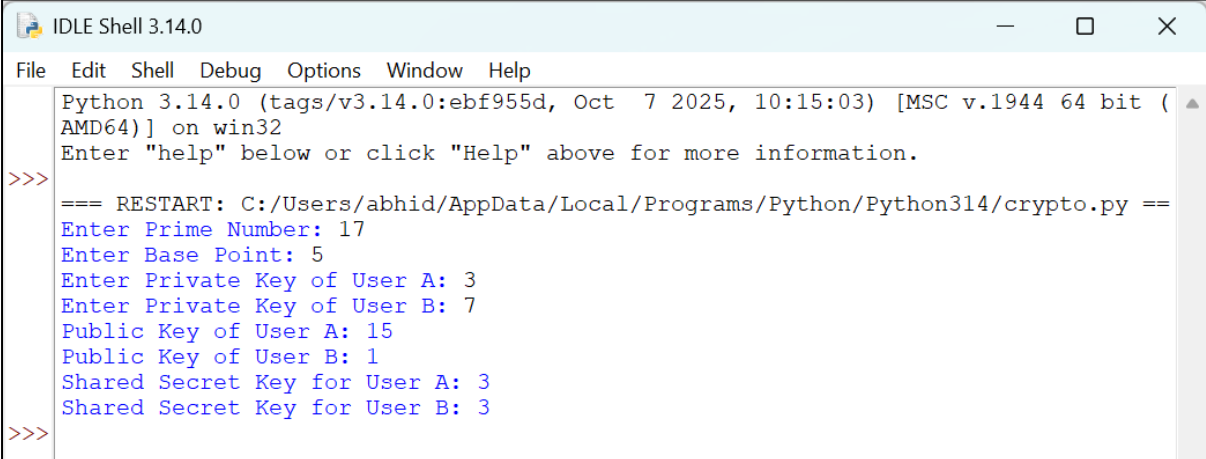
1. Start the program.
2. Enter the prime number p .
3. Enter the base point G .
4. Enter the private keys of User A and User B.
5. Generate the public keys.
6. Exchange the public keys.
7. Compute the shared secret key.
8. Display the public keys and shared secret key.
9. Stop the program.

Code

```
p = int(input("Enter Prime Number: "))
G = int(input("Enter Base Point: "))
a = int(input("Enter Private Key of User A: "))
b = int(input("Enter Private Key of User B: "))
publicA = (a * G) % p
publicB = (b * G) % p
secretA = (a * publicB) % p
```

```
secretB = (b * publicA) % p
print("Public Key of User A:", publicA)
print("Public Key of User B:", publicB)
print("Shared Secret Key for User A:", secretA)
print("Shared Secret Key for User B:", secretB)
```

Output



```
IDLE Shell 3.14.0
File Edit Shell Debug Options Window Help
Python 3.14.0 (tags/v3.14.0:ebf955d, Oct 7 2025, 10:15:03) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/abhid/AppData/Local/Programs/Python/Python314/crypto.py ===
Enter Prime Number: 17
Enter Base Point: 5
Enter Private Key of User A: 3
Enter Private Key of User B: 7
Public Key of User A: 15
Public Key of User B: 1
Shared Secret Key for User A: 3
Shared Secret Key for User B: 3
>>>
```